

Tokenization, Embeddings, and API Keys

1. Tokenization

What is Tokenization?

Tokenization is the process of **splitting text into smaller units** called tokens.

Tokens can be **words**, **characters**, or **subwords** depending on the method used.

Why Tokenize?

- Machines do not understand raw text; they understand numbers.
- Tokenization converts text into a format that models can process and understand.

Example:

Given the text:

"I love AI."

- **Word-level Tokenization:** ["I", "love", "AI", "."]
 - **Character-level Tokenization:** ["I", " ", "l", "o", "v", "e", " ", "A", "I", "."]
 - **Subword Tokenization** (used in models like GPT): ["I", " love", " AI", "."]
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2. Embeddings

What are Embeddings?

Embeddings are **numerical vector representations** of tokens, words, sentences, or even documents.

They capture the **meaning and relationships** between words based on context.

Why Use Embeddings?

- Machines can process and compare numbers, not text.
- Embeddings allow models to understand semantic similarity, context, and meaning.

Example:

Word	Example Embedding (4-D Vector)
cat	`[0.4, 0.2, 0.1, 0.9]`
dog	`[0.5, 0.1, 0.2, 0.8]`
car	`[0.9, 0.5, 0.7, 0.2]`

Notice:

- "cat" and "dog" embeddings are closer than "car", representing their semantic similarity.

Applications:

- Semantic Search
 - Chatbots
 - Recommendations
 - Text Classification
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3. API Keys

What is an API Key?

An **API Key** is a **secret authentication token** that identifies who you are when you interact with an API.

It acts like a **password for programs** to request access to services.

Why Are API Keys Needed?

- To control and secure access.
- To monitor and limit usage.
- To prevent unauthorized users.

Example: OpenAI API Key Usage

python

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```
import openai
```

```
openai.api_key = "sk-XXXXXXXXXXXXXXXXXXXXXXXXXXXX"
```

```
response = openai.ChatCompletion.create(  
    model="gpt-4",  
    messages=[{"role": "user", "content": "Hello!"}]  
)
```

- The "sk-..." string is your **OpenAI API Key**.
- Without the correct key, the service will reject requests.

Important Guidelines:

- **Never share** your API keys publicly.
- **Store securely** using environment variables (.env files) or secret managers.
- **Rotate** keys periodically for enhanced security.